# Finding Similar Items 2

Introduction to Data Mining

Seoul National University

Spring 2026

# Announcements

- **Assignments:**
  - HW1 is posted (deadline: April 9).

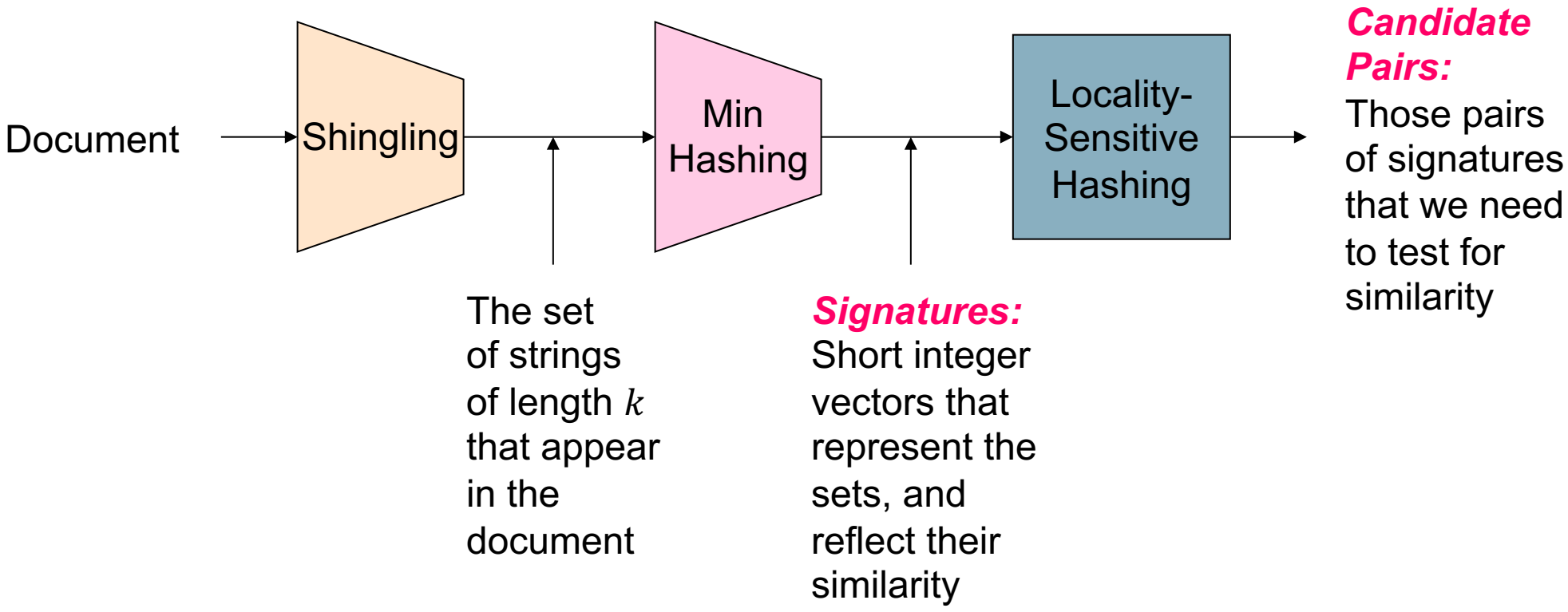
# Recap: Shingling and Minhashing

- $k$-shingles (document as a set)
- Jaccard similarity:
  - $\mathrm{sim}(S, T) = |S \cap T| / |S \cup T|$
- Minhashing:
  - Pick a random permutation of rows.
  - The minhash value is the first row that has 1.
  - E.g., $h(S_1) = 0$, $h(S_2) = 2$, and $h(S_3) = 1$.
  - $\Pr(\text{minhash is same}) =$ Jaccard similarity.
- How to compute minhash signatures.

| *Row* | *Element* | $S_1$ | $S_2$ | $S_3$ |
|---|---|---|---|---|
| 0 | "The plane" | 1 | 0 | 0 |
| 1 | "The cow j" | 0 | 0 | 1 |
| 2 | "Alice and" | 0 | 1 | 0 |
| 3 | "Roses are" | 1 | 0 | 1 |

| *Row* | $S_1$ | $S_2$ | $S_3$ |
|---|---|---|---|
| 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 |
| 3 | 1 | 0 | 1 |
| 2 | 0 | 1 | 0 |

# Recap: The Big Picture




Source: Stanford CS246 (2018)

# Outline

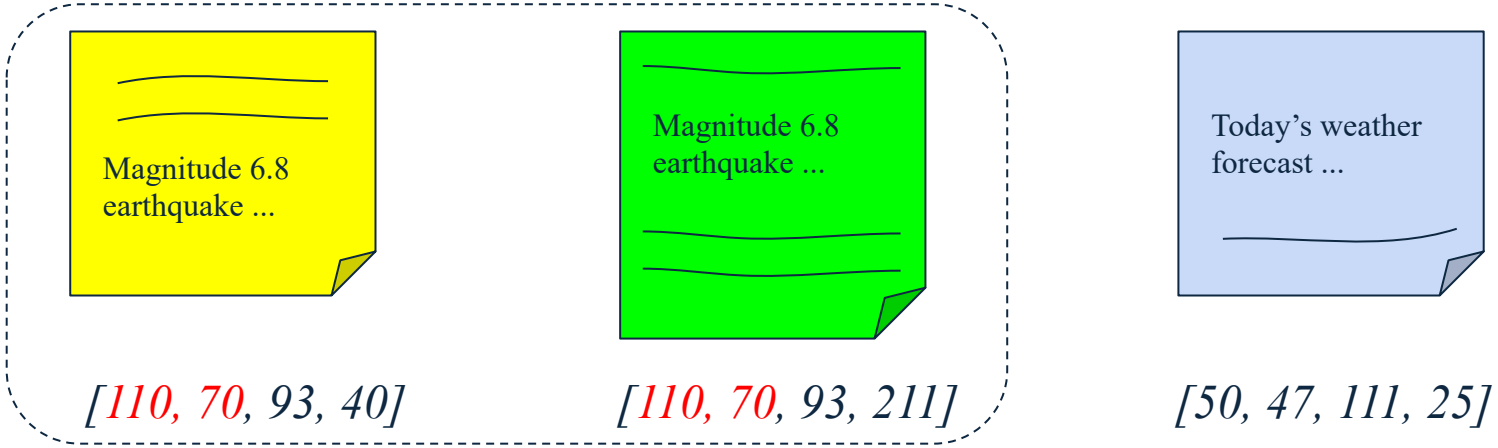
1. **<u>Locality-Sensitive Hashing (LSH)</u>**
2. S-Curves
3. LSH Families
4. LSH with Other Distance Measures

# Locality-Sensitive Hashing

- The number of document pairs is still large after minhashing.
  - Minhashing does not reduce document pairs to compare.
  - What if we have 1M documents with 1μs for each comparison?
  - 6 days to compute all similarities.
- **Locality-sensitive hashing (LSH)** reduces # of candidate pairs.
  - Allows us to focus on pairs that are likely to be similar.

# From Signatures to Buckets

- **General approach:** Hash minhashed signatures several times.
  - We know similar documents have more shared signatures.
  - Use only a subset of the signatures for each hash function.
- Any pair hashed to the same bucket becomes a candidate.
  - There will be false positives and false negatives though.

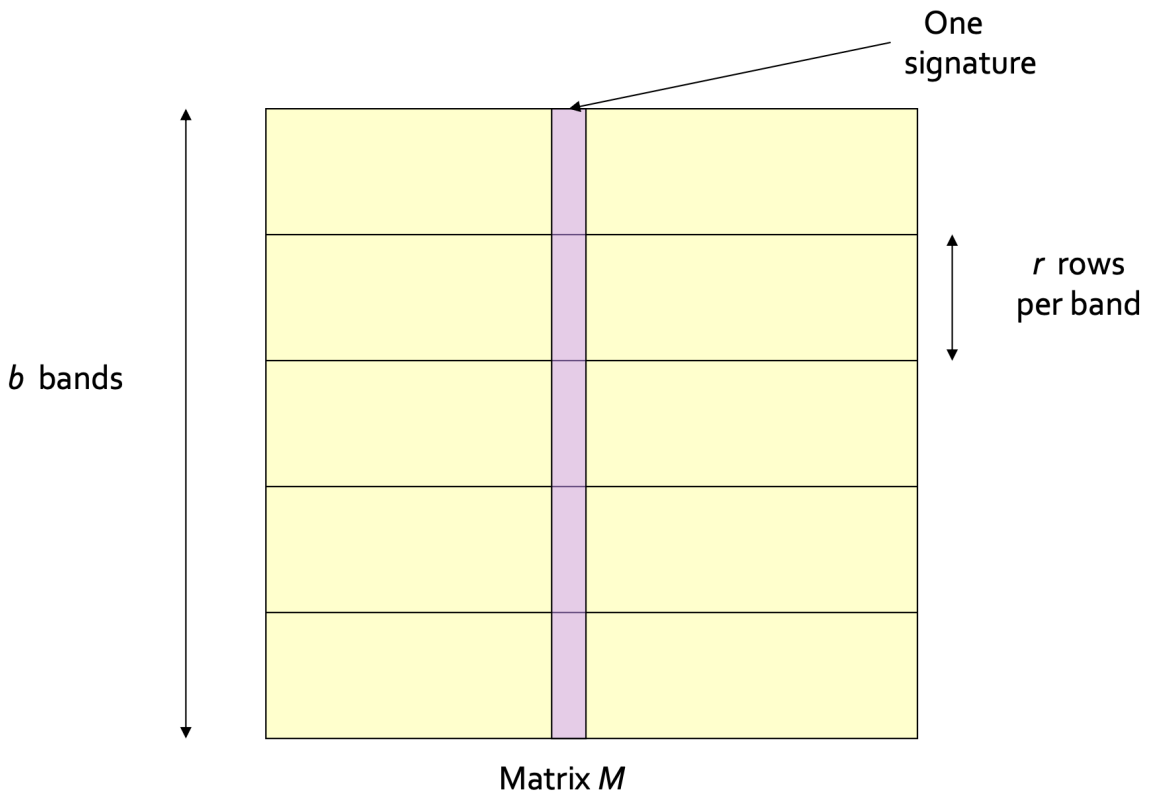


# Partition into Bands

- **Trick:** Divide the signature matrix $M$ into $b$ **bands** of $r$ rows.
  - For each band, hash each column to a hash table with $k$ buckets.
    - Make $k$ as large as possible to reduce collisions.
    - Use a different hash table for each band.
  - Becomes a candidate pair if hashed to the same bucket for $\geq 1$ band.
  - Tune $b$ and $r$ to catch most similar pairs, but few non-similar pairs.

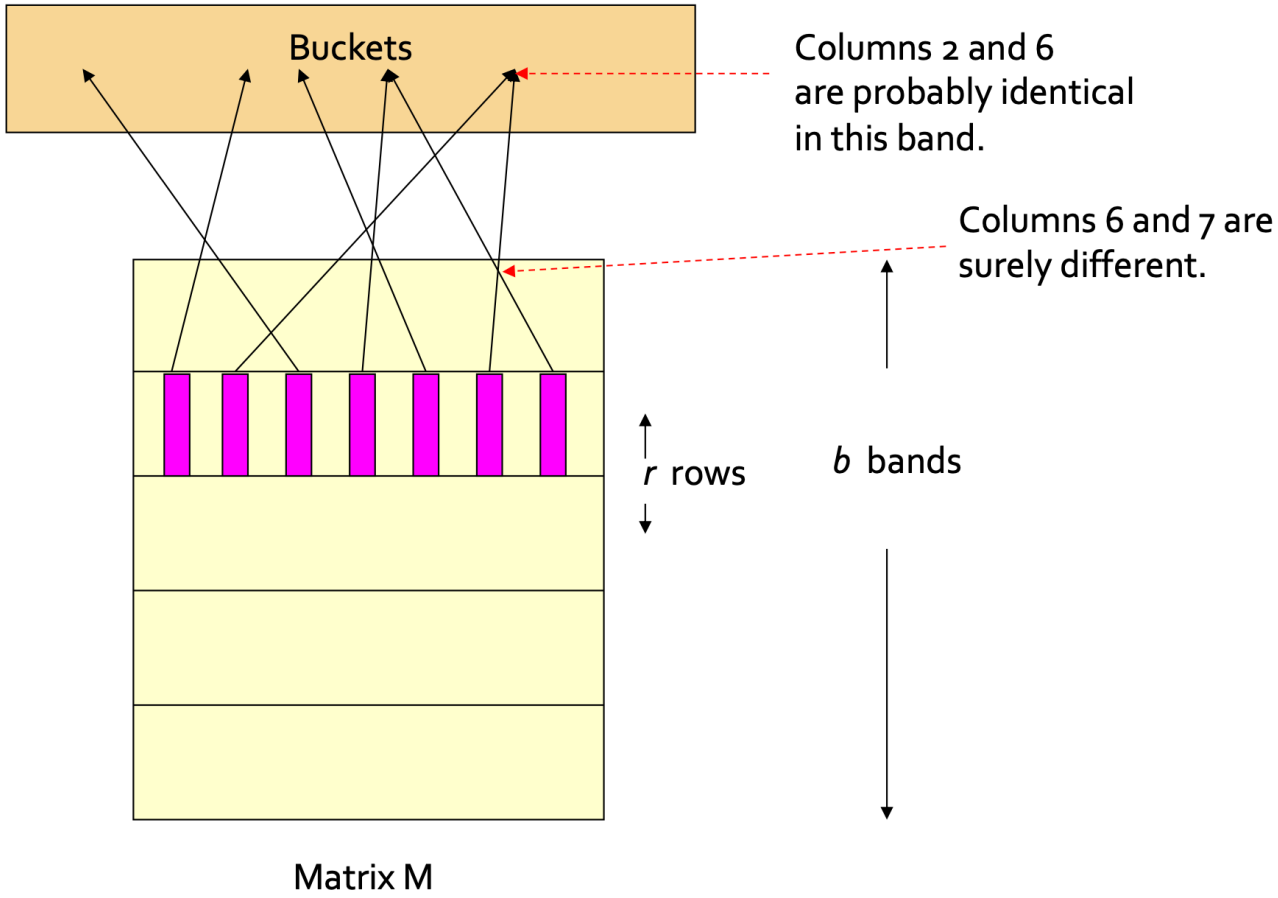
# Partition into Bands

- **Trick:** Divide the signature matrix $M$ into $b$ **bands** of $r$ rows.




Source: Stanford CS246 (2022)

# Hash Functions to One Bucket




Source: Stanford CS246 (2022)

# Example: Bands

- Suppose a signature matrix of size $100 \times 100{,}000$.
  - 100 minhashed signatures
  - 100,000 documents to compare
- **Goal:** Find all 80%-similar pairs.
  - 5,000,000,000 pairs of signatures can take a while to compare.
  - Let's apply LSH with $b = 20$ bands of size $r = 5$.

# Example: False Positives

- Suppose that $C_1$ and $C_2$ are 40% similar (i.e., negative pair).
- **False positives:** Less than 1/5th of all 40%-similar pairs.
  - Probability identical in any one particular band: $(0.4)^5 = 0.01$.
  - Probability identical in $\geq 1$ of 20 bands: $1 - (1 - 0.01)^{20} < 0.2$.
- We can remove more than 80% of all 40%-similar pairs!

# Example: False Negatives

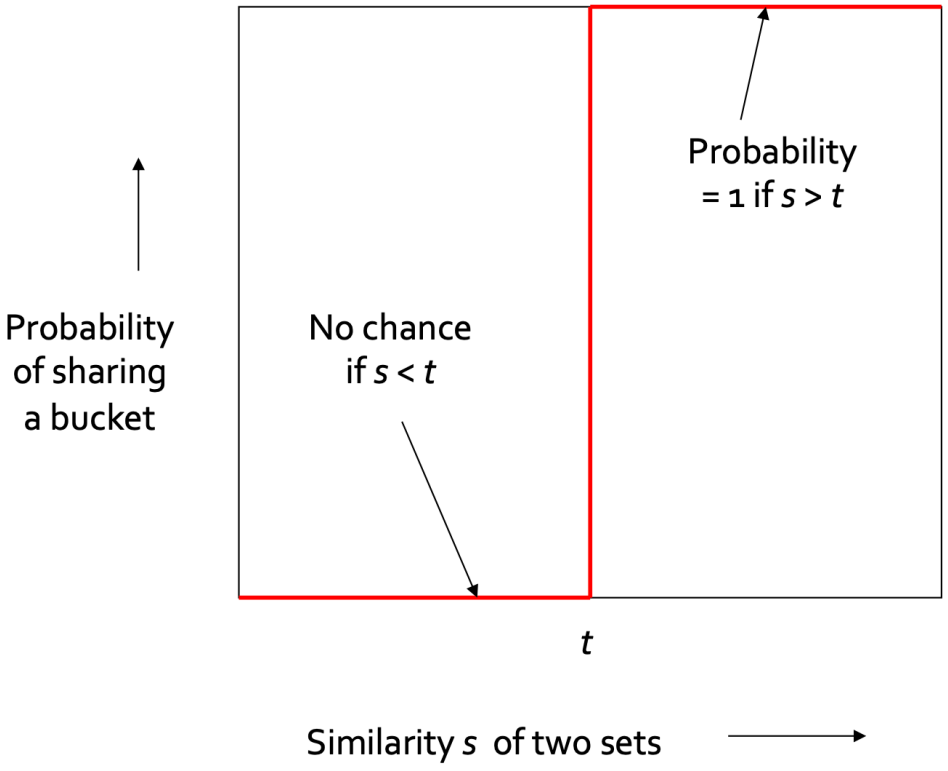
- Suppose that $C_1$ and $C_2$ are 80% similar (i.e., positive pair).
- **False negatives:** About 1/3000th of all 80%-similar pairs.
  - Probability identical in any one particular band: $(0.8)^5 = 0.328$.
  - Probability not similar in all 20 bands: $(1 - 0.328)^{20} = 0.00035$.
- We can find almost all positive pairs even with LSH!

# Outline

1. Locality-Sensitive Hashing (LSH)
2. **<u>S-Curves</u>**
3. LSH Families
4. LSH with Other Distance Measures

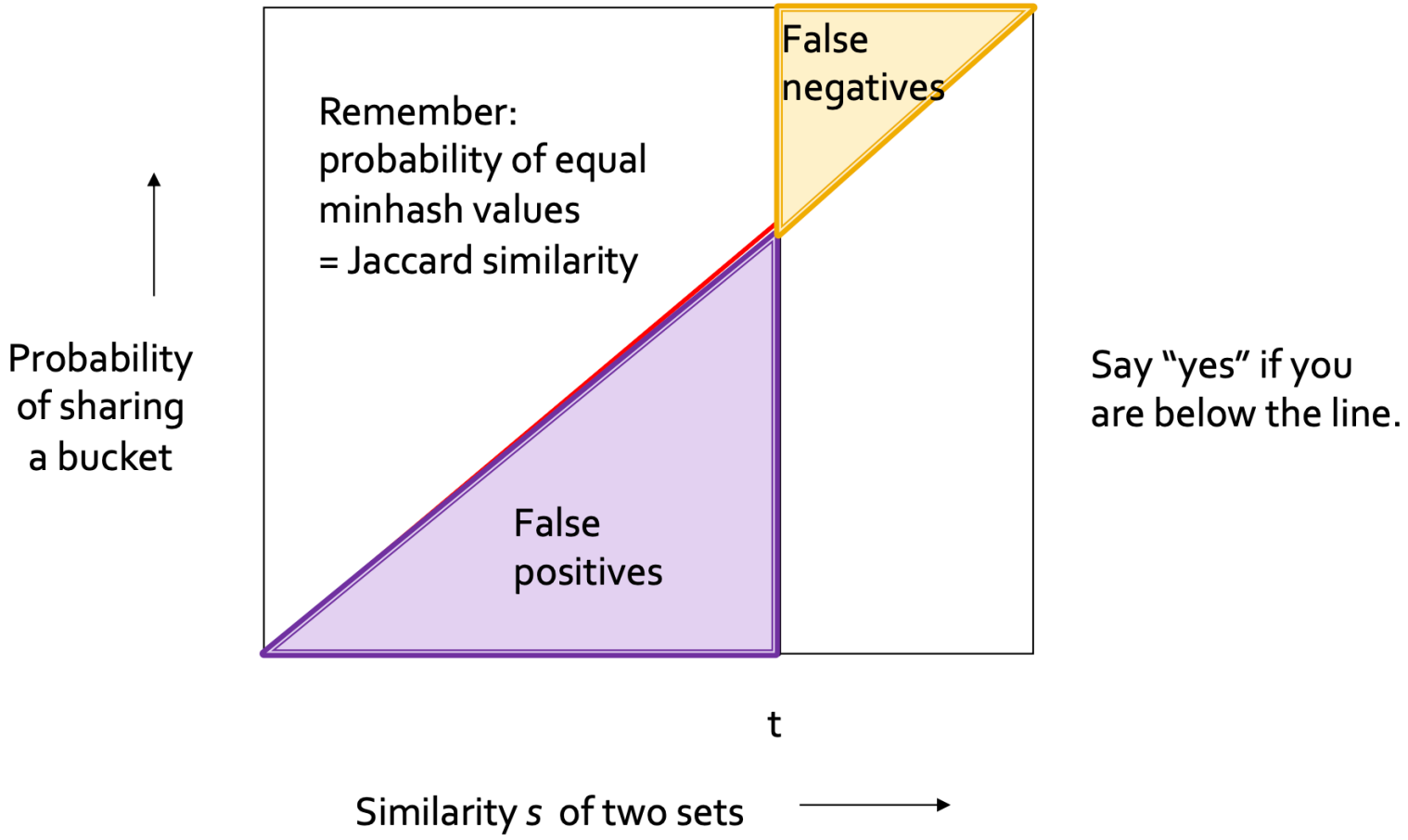
# What We Want from LSH

- Let $s$ be a **similarity**, and $t$ be a **similarity threshold** for the query.
- What we want from LSH is:




Source: Stanford CS246 (2022)

# Errors when $r = 1$ and $b = 1$

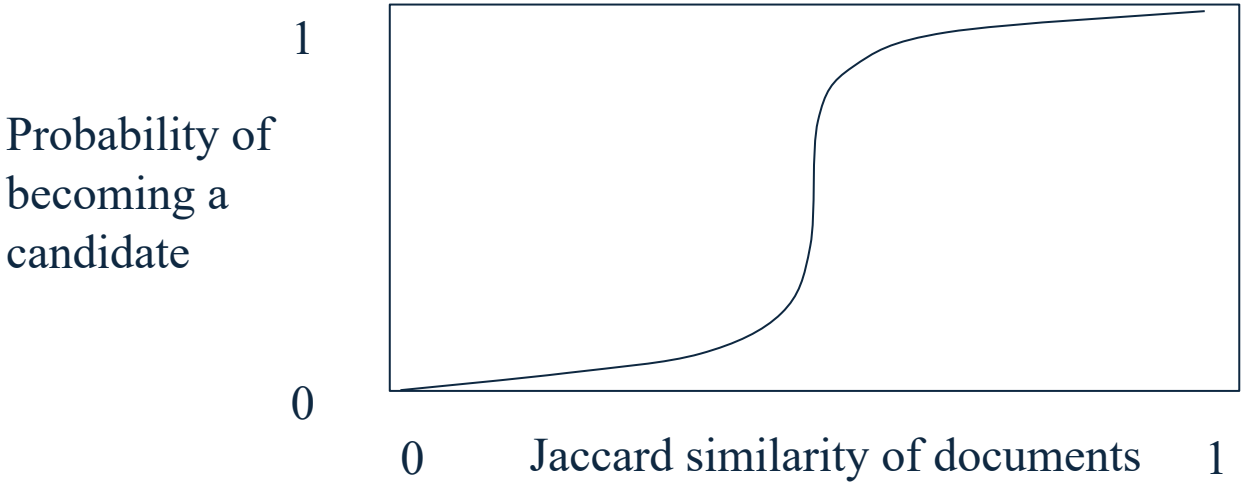



Say "yes" if you are below the line.

Source: Stanford CS246 (2022)

# S-Curve

- LSH makes an **S-curve** for a document pair with similarity $s$.
  - $\Pr(\text{signatures agree in all rows on one band}) = s^r$
  - $\Pr(\text{signatures do not agree in at least one row of one band}) = 1 - s^r$
  - $\Pr(\text{signatures do not agree in at least one row of each band}) = (1 - s^r)^b$
  - $\Pr(\text{signatures agree in all rows of at least one band}) = 1 - (1 - s^r)^b$




| $s$ | $1-(1-s^r)^b$ |
|---|---|
| .2 | .006 |
| .4 | .186 |
| .6 | .802 |
| .8 | .9996 |

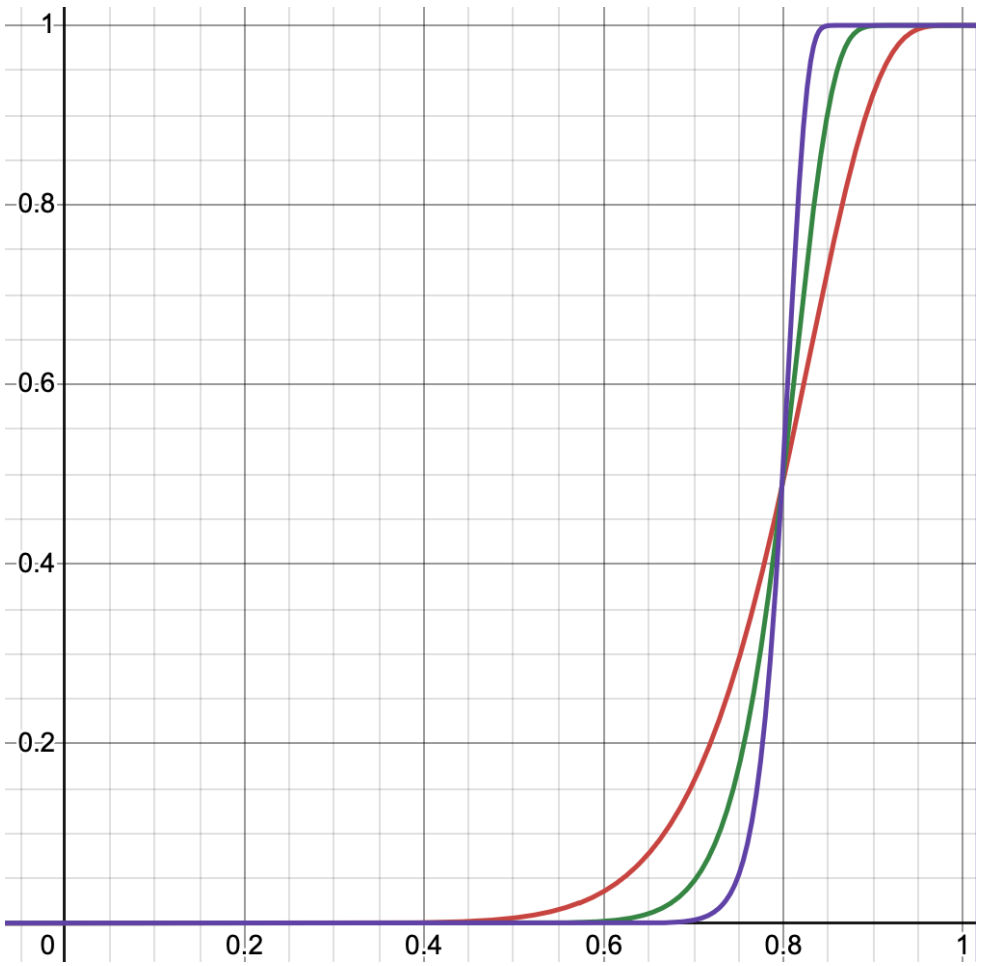
$b = 20, r = 5$

# S-Curve: Choosing $r$ and $b$

- The values of $r$ and $b$ determine the shape of the S-curve.
  - Constraint: $r \times b$ equals to the number of signatures $m$.
- Higher $m$ makes the curve closer to the step function.
- The balance between $r$ and $b$ decide the center of curve.

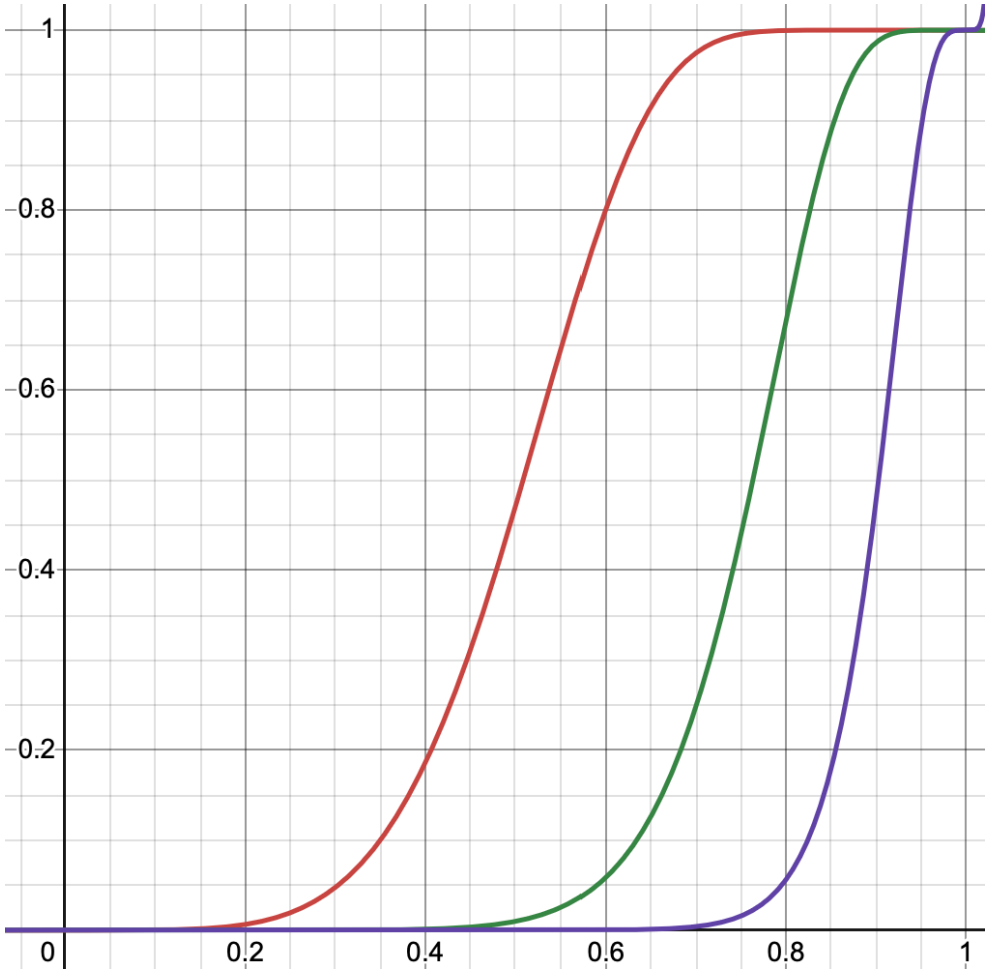
# Changing Shape with $m$

- Suppose that we consider 0.8 as the similarity threshold.
- We visualize three curves with:
  - Red: $r = 10$ and $b = 6$
  - Green: $r = 20$ and $b = 60$
  - Purple: $r = 40$ and $b = 5500$
- Higher $m$ makes higher accuracy.



# Changing Shape with $r$ and $b$

- With same $m$, the values of $r$ and $b$ determines the center of curve.
- We visualize three curves with:
  - Red: $r = 5$ and $b = 20$
  - Green: $r = 10$ and $b = 10$
  - Purple: $r = 20$ and $b = 5$
- No answer; they have different false positives and negatives.



# Finding the Center of Curve

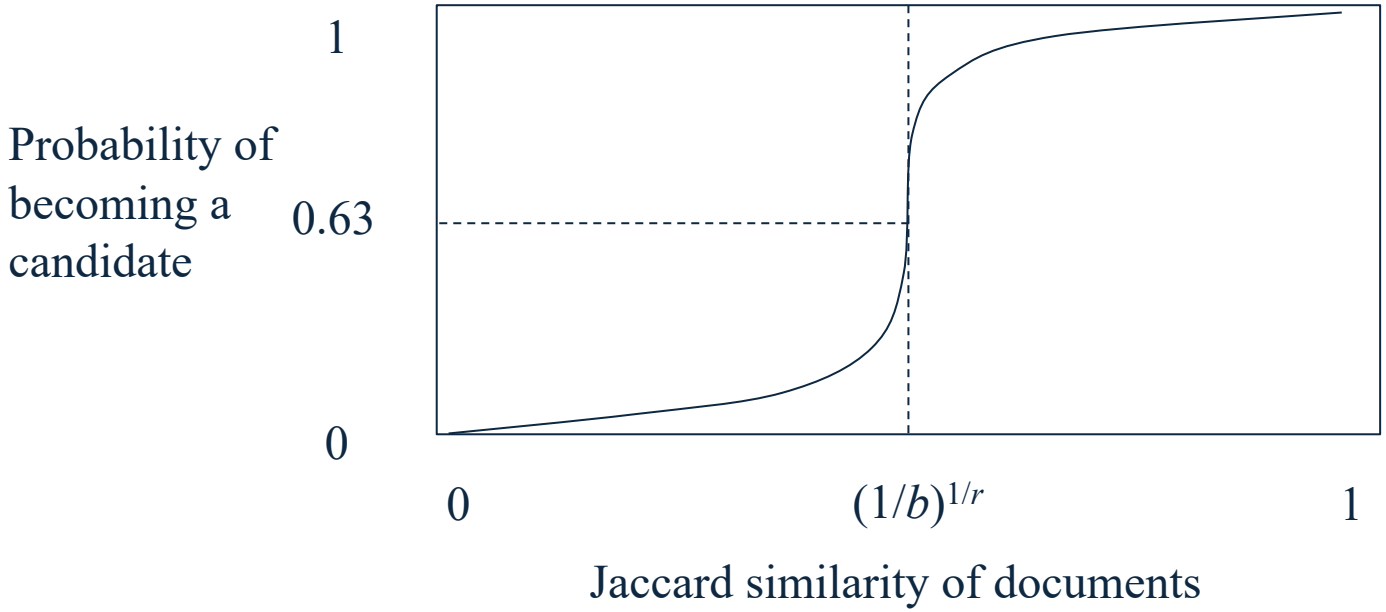
- We can't draw the curve for every combination of $r$ and $b$.
- Let's approximate the equation:

  $y = 1 - (1 - s^r)^b$

  $\approx 1 - e^{-bs^r}$ if $0 < s < 1$ and $r$ is sufficiently large.
- The curve becomes steepest (i.e., "center") if $s \approx (1/b)^{1/r}$.
  - Let's use this center to approximate the shape of curve.

# Finding the Center of Curve

- **Note:** The curve passes $((1/b)^{1/r}, 0.63)$ regardless of $r$ and $b$.



# Combining the Techniques

- Pick $k$ and construct a set of $k$-shingles from each document.
- Pick length $n$ for minhash signatures and compute signatures.
- Pick similarity threshold $t$ and then # of bands $b$ and # of rows $r$.
  - If $(1/b)^{1/r} > t$, there will be fewer false positives.
  - If $(1/b)^{1/r} < t$, there will be fewer false negatives.
- Construct candidate pairs and compare their signatures using $t$.

# Outline

1. Locality-Sensitive Hashing (LSH)
2. S-Curves
3. **<u>LSH Families</u>**
4. LSH with Other Distance Measures

# Distance Measures

- Generalized LSH is based on the **distance** between points.
  - Jaccard similarity is not a distance.
  - Jaccard distance = 1 – Jaccard similarity is a distance.
- A function $d$ is a **distance measure** if
  - It takes two points in the space and returns a real number satisfying:
    - $d(x, y) \geq 0$ (non-negative)
    - $d(x, y) = 0$ iff $x = y$ (distances are positive except for $x = y$)
    - $d(x, y) = d(y, x)$ (symmetric)
    - $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality)

# LSH Families

- Let $d_1 < d_2$ be two distances according to a distance function $d$.
- A family $F$ of functions is said to be $\boldsymbol{(d_1, d_2, p_1, p_2)}$**-sensitive** if
  - For every $f \in F$ and points $x$ and $y$:

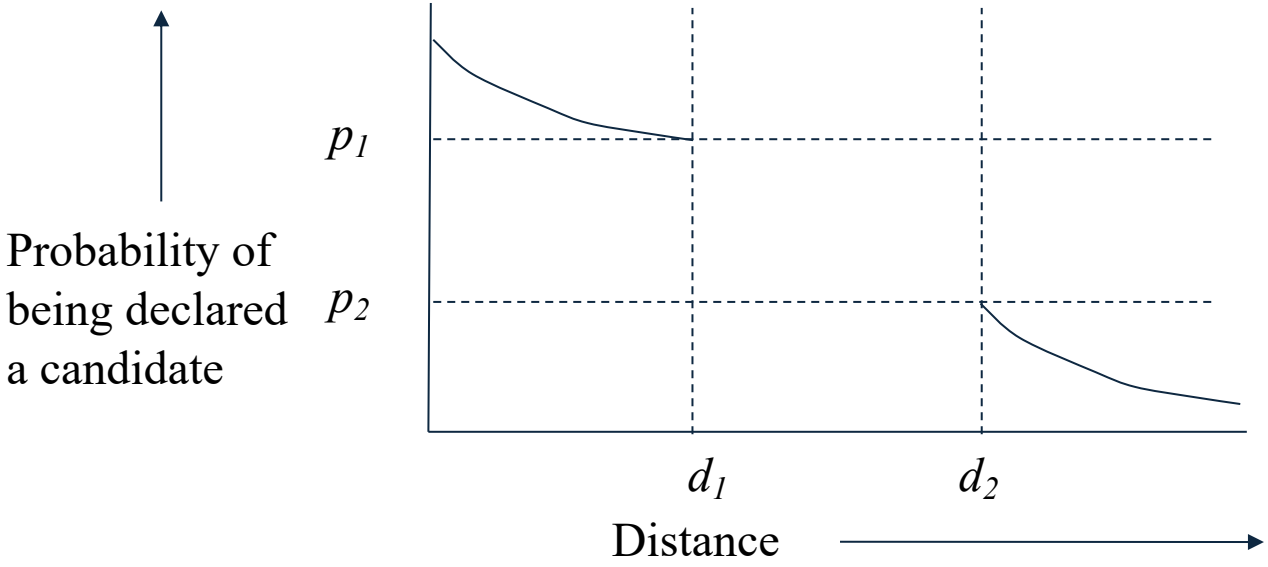
    If $d(x, y) \leq d_1$, then the probability that $f(x) = f(y)$ is at least $p_1$.

    If $d(x, y) \geq d_2$, then the probability that $f(x) = f(y)$ is at most $p_2$.
- The values $(d_1, d_2, p_1, p_2)$ are related to the shape of the curve.

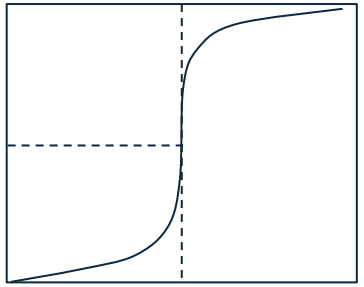
# Meanings of Sensitivity Values

- If $F$ is $(d_1, d_2, p_1, p_2)$-sensitive, then

  $F$ is better with higher $d_1$, lower $d_2$, higher $p_1$, and lower $p_2$.




Reflection of the S-curve across the $y$-axis.



# LSH Families for Jaccard Similarity

- Minhashing gives a $(d_1, d_2, 1 - d_1, 1 - d_2)$-sensitive family.
  - For any $d_1$ and $d_2$ such that $0 \leq d_1 < d_2 \leq 1$.
  - Here we do not consider the banding technique (i.e., $b = r = 1$).
- **Proof:**
  - Let $d$ be the Jaccard distance $(= 1 - \mathrm{sim}(x, y))$.
  - Then, $d(x, y) \leq d_1$

    $\Rightarrow 1 - \mathrm{sim}(x, y) \leq d_1$

    $\Rightarrow 1 - d_1 \leq \mathrm{sim}(x, y)$

    $\Rightarrow 1 - d_1 \leq \Pr(\mathrm{minhash}(x) = \mathrm{minhash}(y))$.
  - Similar argument applies to $d_2$.

# Amplifying an LSH Family

- The **banding technique** carries over to this general setting.
- We create a new LSH family $F'$ from $F$ such that:
  - Each member of $F'$ consists of $n$ members of $F$.
  - By an **AND construction** like "rows in a band."
  - And an **OR construction** like "many bands."
- **Goal:** Improve $F$ by making $d_1\uparrow$, $d_2\downarrow$, $p_1\uparrow$, and $p_2\downarrow$.

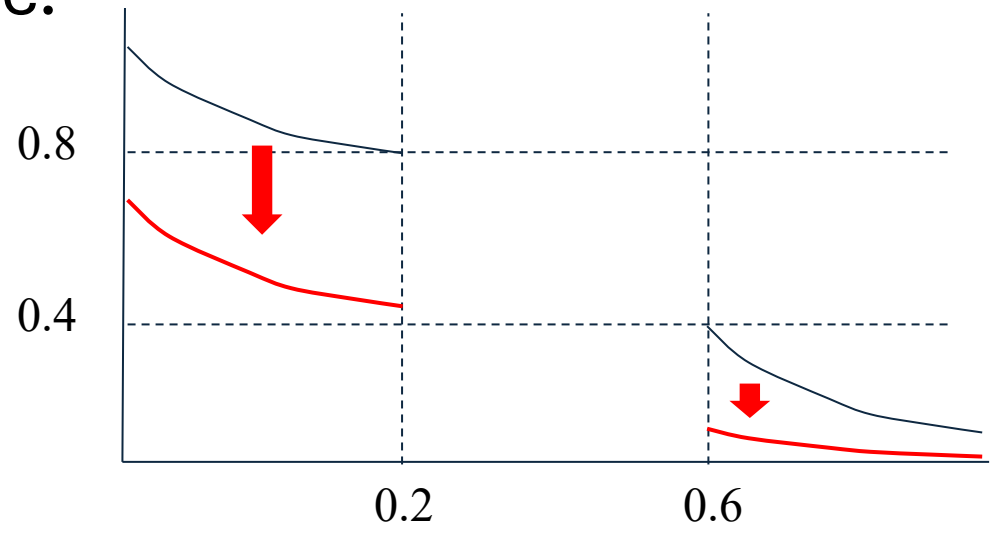
# AND Construction

- If the following are true:
  - $f' \in F'$ is constructed from $\{f_1, f_2, \dots, f_r\}$ of $F$.
  - $f'(x) = f'(y)$ if and only if $f_i(x) = f_i(y)$ **for all** $i = 1, 2, \dots, r$.
- Then, $F'$ is $(d_1, d_2, p_1^r, p_2^r)$-sensitive.

F is a (0.2, 0.6, 0.8, 0.4)-sensitive family

If $r = 4$, $0.8^4 =$ $0.4^4 =$

F' is (0.2, 0.6, 0.4096, 0.0256)-sensitive



# OR Construction

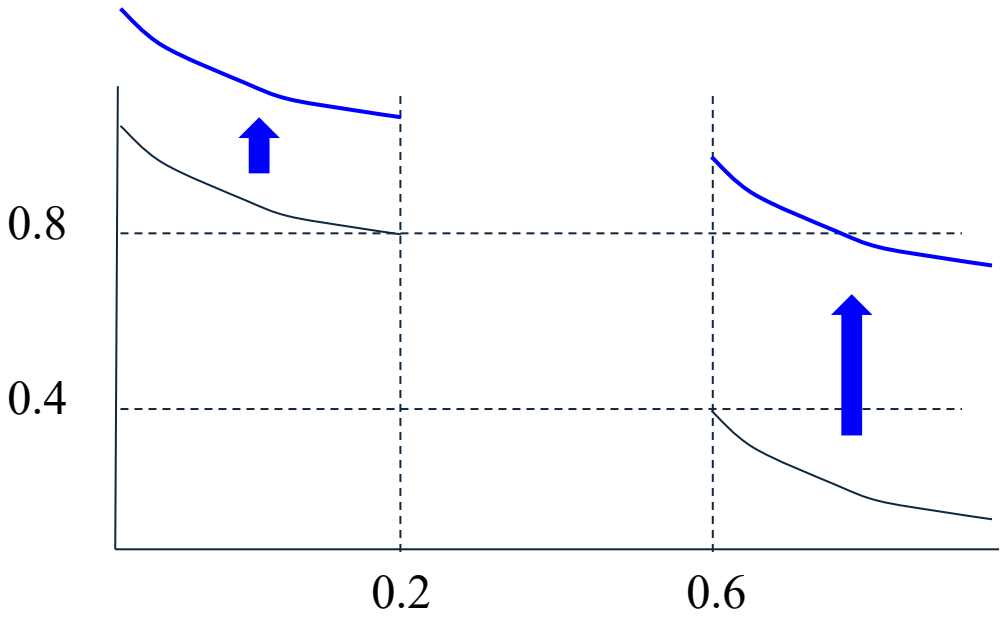
- If the following are true:
  - $f' \in F'$ is constructed from $\{f_1, f_2, \dots, f_b\}$ of $F$.
  - $f'(x) = f'(y)$ if and only if $f_i(x) = f_i(y)$ **for at least one** $i = 1, 2, \dots, b$.
- Then, $F'$ is $(d_1, d_2, 1-(1-p_1)^b, 1-(1-p_2)^b)$-sensitive.

F is a (0.2, 0.6, 0.8, 0.4)-sensitive family

If $b = 4$,

F' is (0.2, 0.6, 0.9984, 0.8704)-sensitive

$1-(1-0.8)^4 =$ $1-(1-0.4)^4 =$

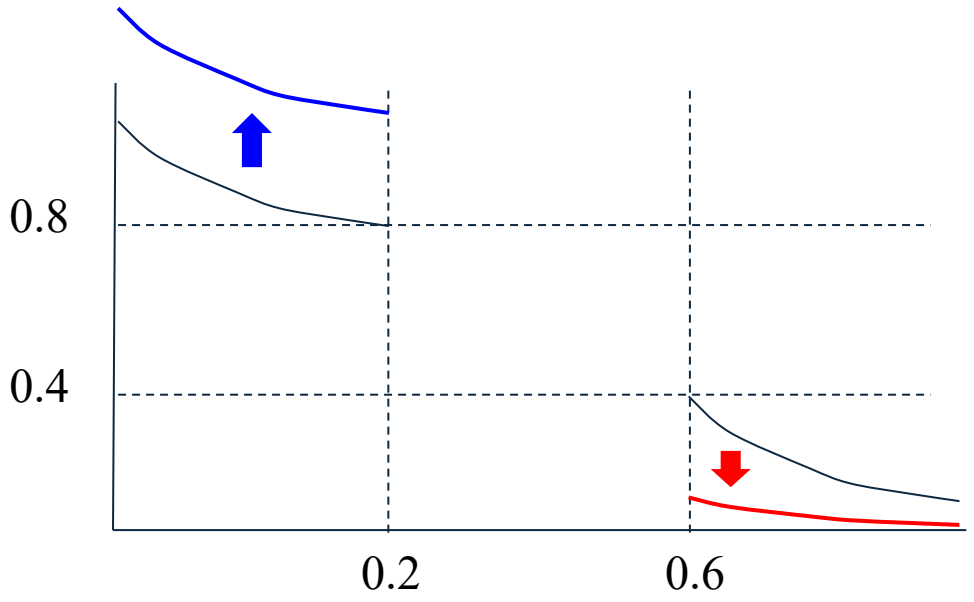


# AND-OR Composition

- Let's combine an AND and an OR construction as in minhashing.
  - In the right order: AND → OR, not the opposite!
- Each of the two probabilities $p$ is transformed into $1-(1-p^r)^b$.
  - The S-curve studied before!

F is a (0.2, 0.6, 0.8, 0.4)-sensitive family

If $r = b = 4$,
F' is (0.2, 0.6, 0.8785, 0.0985)-sensitive

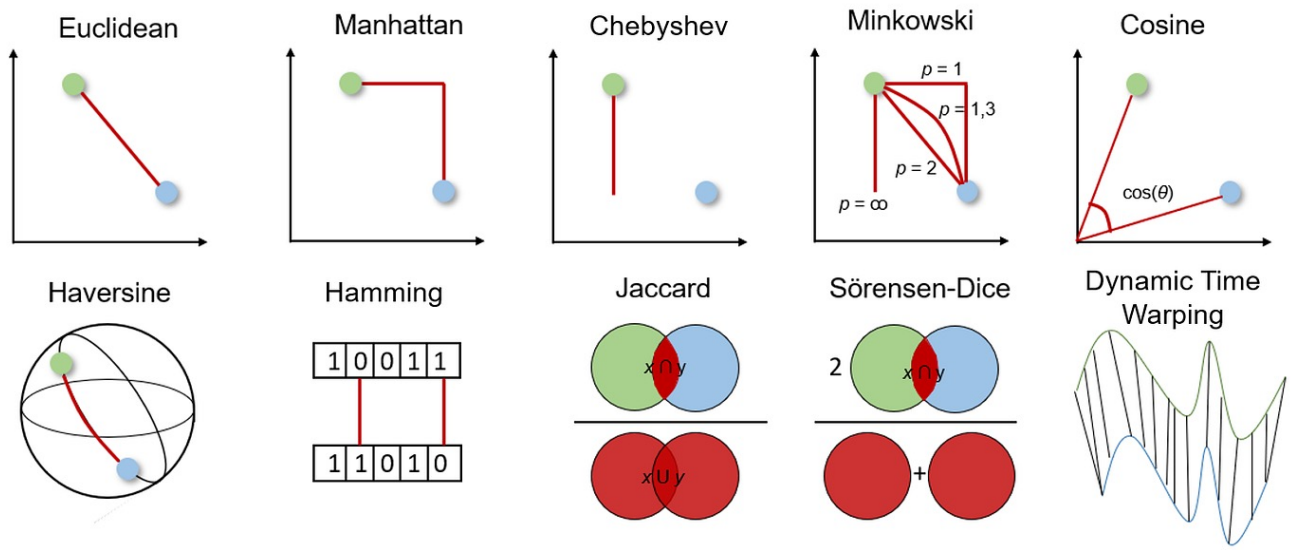


# Outline

1. Locality-Sensitive Hashing (LSH)
2. S-Curves
3. LSH Families
4. **<u>LSH with Other Distance Measures</u>**

# LSH for Other Distance Measures

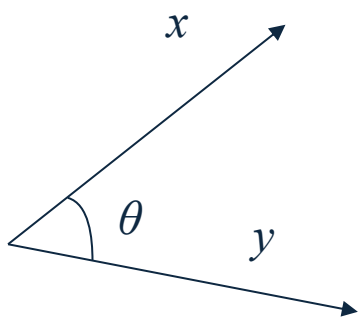
- Let's study the **angular distance** between real-valued vectors.
- See textbook for other distance measures:
  - Euclidean distance, Hamming distance, Edit distance, etc.




Source: Medium

# Angular Distance

- Think of a point as a vector from the origin to its location.
- Two points' vectors create an angle $\theta$.
- Then, $\theta/\pi$ is the **angular distance** between the vectors.
- Normally computed from the **cosine similarity** $\cos(\theta)$.

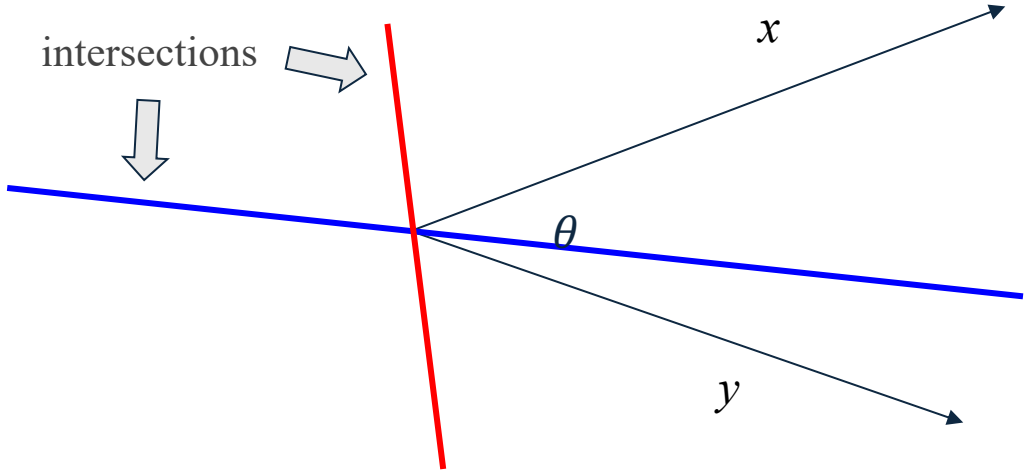



$$cos(\theta) = \frac{x \cdot y}{||x|| \cdot ||y||}$$

Cosine similarity

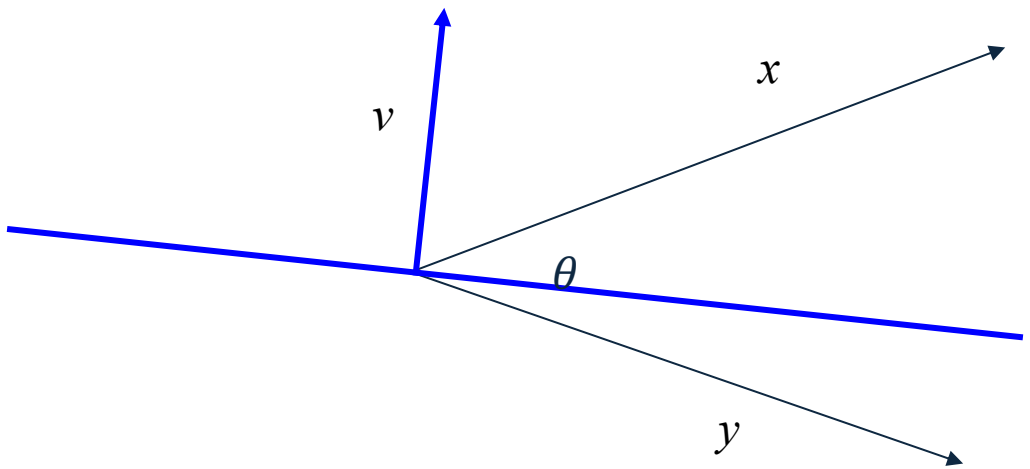
# Random Hyperplanes

- **Random hyperplanes:** Technique analogous to minhashing.
- Two vectors $x$ and $y$ that have an angle $\theta$ define a plane.
- All hyperplanes through the origin can be categorized into
  - Red hyperplane where $x$ and $y$ are on the same side.
  - Blue hyperplane where $x$ and $y$ are on different sides .



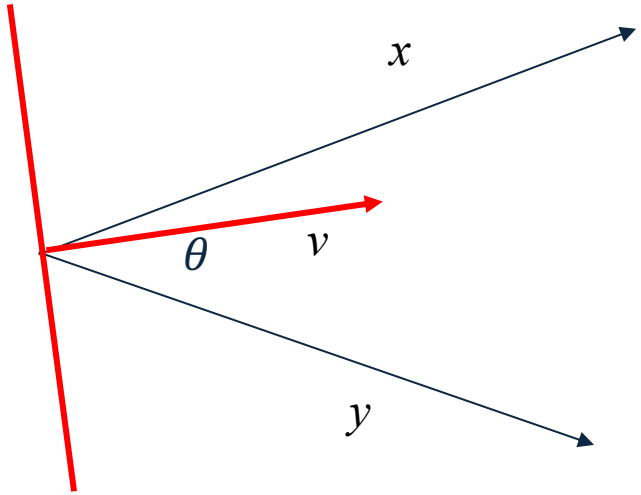
# Blue Hyperplane Case

- Consider a vector $v$ that is normal to a blue-type hyperplane.
- Then, $v \cdot x$ and $v \cdot y$ will have different signs.
  - Since $x$ and $y$ are on different sides.
- Even if $v$ extends in opposite direction, signs are still different.



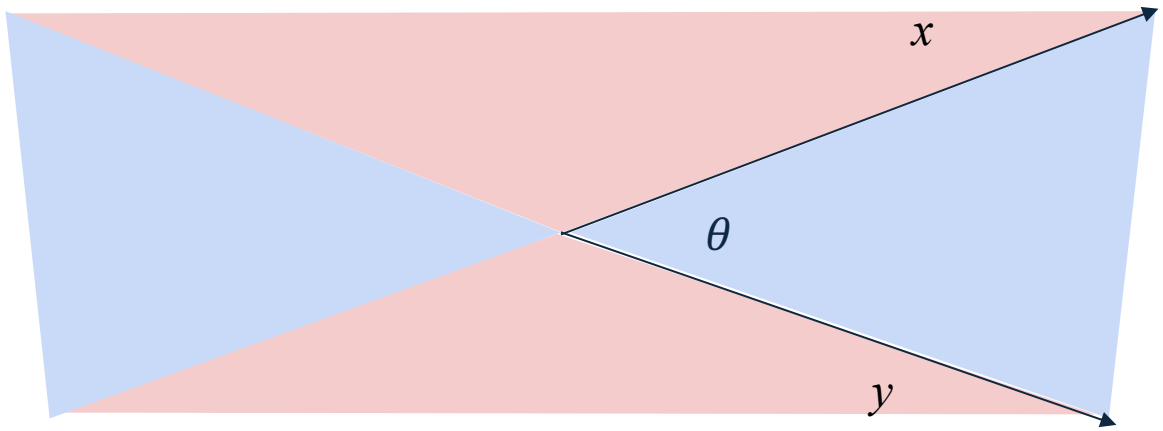
# Red Hyperplane Case

- Suppose that $v$ is normal to a red-type hyperplane.
- Then, $v \cdot x$ and $v \cdot y$ have the same sign.



# Blue and Red Probabilities

- Suppose that $v$ is randomly chosen.
- The probability that a hyperplane will be red-type: $1 - \theta/\pi$.



# Random Hyperplanes

- If the following are true:
  - Hash function $f \in F$ is built from a randomly-chosen vector $v$.
  - Given vectors $x$ and $y$, $f(x) = f(y)$ iff $v \cdot x$ and $v \cdot y$ have the same sign.
    - Red-type hyperplane.
- Then, $F$ is $(d_1, d_2, 1 - d_1/\pi, 1 - d_2/\pi)$-sensitive.
- We can amplify $F$ as we wish, just like the minhash-based family.

# Summary

1. Locality-Sensitive Hashing (LSH)
   - Banding technique
2. S-Curves
   - Choosing the values of $r$ and $b$
3. LSH Families
   - $(d_1, d_2, p_1, p_2)$-sensitivity
   - AND and OR constructions
4. LSH with Other Distance Measures
   - Random hyperplanes